What is LVM?

In simple terms, **LVM (Logical Volume Manager)** is a tool in Linux that helps you manage your computer's storage. It lets you create, resize, and organize your storage space without being stuck with the size of your hard drives or partitions.

Imagine LVM as a flexible **storage organizer**. You can move, resize, and even combine storage from different hard drives to create one big storage space that's easy to manage. It helps you make the most of your storage and avoid running out of space.

Don't worry if this sounds a bit complicated right now. As you read on, we'll explain how volumes work and how LVM fits into the picture.

What are volumes in linux ?

volumes in Linux are like containers that hold your data. They represent storage space on your computer or server, and you can use them to store files and programs.
These volumes can be physical (like an SSD or HDD), partitioned sections of a physical disk, or even logical/virtual disks managed by a tool called **LVM (Logical Volume Manager)**.

Types of Volumes

1. Physical Volume (PV)

- A **physical volume** is the actual storage device, like an **HDD (Hard Disk Drive) or SSD (Solid State Drive)**.

  Example: Your laptop's SSD or an external hard drive.

📌 **Key Point:** Every volume must exist on a physical storage device.

2. Volume Group (VG)

- A storage pool made from multiple physical volumes.

  Example: Combining an SSD[10 GB] and HDD [10 GB] in one group .

📌 **Key Point**: VGs merge multiple devices into one.

3. Logical Volume (LV) - Managed by LVM

- A flexible storage unit inside a volume group, easy to resize or move.

  Example: Creating a 10GB Logical Volume (LV) from a 20GB Volume Group (VG)

📌 **Key Point**: LVs allow for dynamic storage management, including resizing and moving.

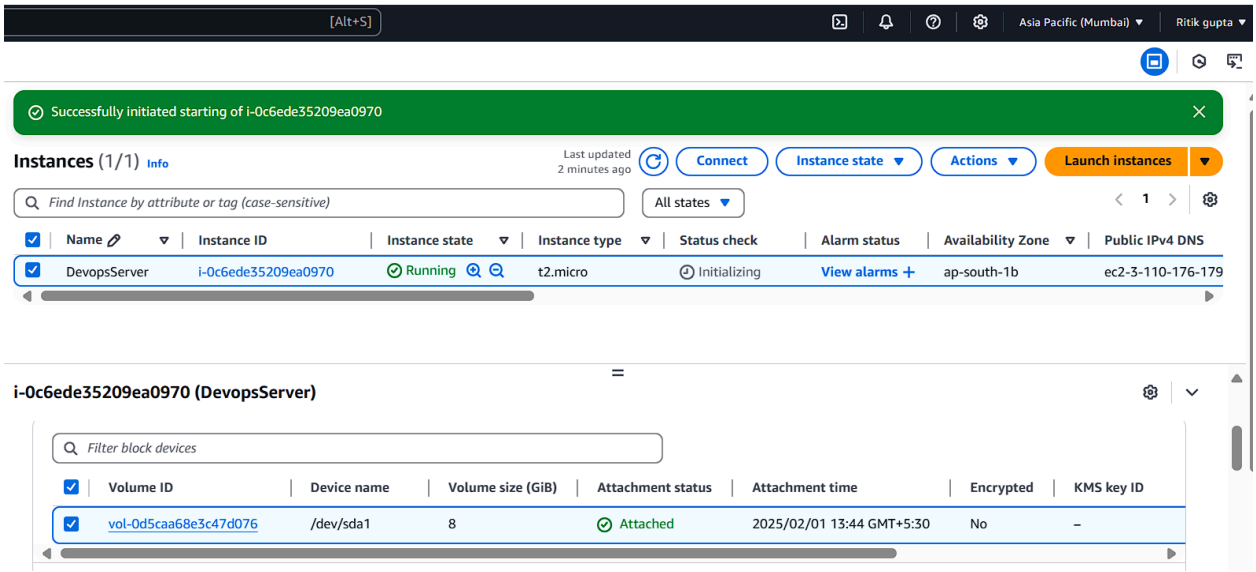
What is AWS EBS?

AWS **EBS (Elastic Block Store)** is a block storage service that provides persistent storage for Amazon EC2 instances. It behaves like a physical volume (PV) in Linux, offering block-level storage that can be attached to instances.

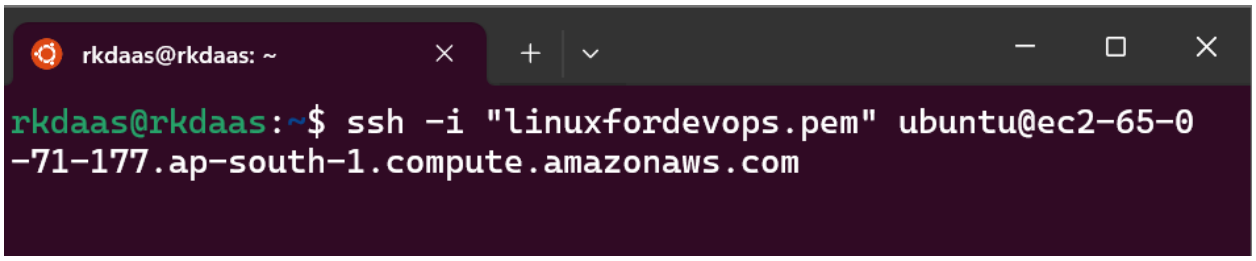
📌 **Key Point:** AWS EBS is **not an LV**, but it is **used as a PV in LVM**.
📌 You can create **Logical Volumes (LVs)** inside AWS EBS by using **LVM** for more flexibility in managing your storage.

Now, let's set up LVM on top of AWS EBS

I already have an EC2 instance running, but you can launch one yourself—it's a simple process. By default, AWS provides an **8GB EBS volume** with your instance unless you customize the storage settings during creation.



Connect to Remote Server via SSH



Now Let’s understand some Important commands

`**1- Lsblk** -This command in Linux shows a clear list of all storage devices and their partitions. Think of it as a map of your system's disks—it tells you what storage is available, how it's divided, and where it's mounted.

```
rkdaas@rkdaas: ~
ubuntu@ip-172-31-5-208:~$ lsblk
NAME     MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0      7:0    0 26.3M  1 loop /snap/amazon-ssm-agent/988
1
loop1      7:1    0 73.9M  1 loop /snap/core22/1722
loop2      7:2    0 44.4M  1 loop /snap/snapd/23545
loop3      7:3    0 73.9M  1 loop /snap/core22/1748
xvda     202:0    0    8G  0 disk
├─xvda1  202:1    0    7G  0 part /
├─xvda14 202:14   0    4M  0 part
├─xvda15 202:15   0  106M  0 part /boot/efi
└─xvda16 259:0    0  913M  0 part /boot
```

Breaking it Down

- **NAME** → The disk name (`xvda`)
- **SIZE** → Total size of the disk
- **TYPE** → Whether it's a disk or partition
- **MOUNT POINT** → If mounted, it shows where (e.g., `/` or `/mnt/data`)

📌 If a volume is **attached** but **not mounted**, the "MOUNTPOINT" column will be **empty**.
📌 If a volume is **mounted**, it will show the directory where it is accessible.

Difference between Attached and mounted

1️⃣ **Attached:**

- The storage (like an EBS volume) is connected to the system.
- It's not yet available for use or accessible by the OS.
- **Key Point:** Just connected, but not ready to store data yet.

2️⃣ **Mounted:**

- The storage is linked to a specific folder (mount point), making it usable.
- You can access it and store data in that folder.

- **Key Point:** Mounted means it’s now accessible and ready for use.

📌 **Example in AWS:**

- When you attach an EBS volume to an EC2 instance, it's connected, but not usable until you mount it to a folder (e.g., `/mnt/data`).
- **Key Point:** You can't store data on the volume until it’s mounted.

2- Df -h - This command helps monitor your filesystem(volumes) space usage and provides a warning when storage is running low, so you can take action before running out of space.

```
ubuntu@ip-172-31-5-208:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       6.8G  1.8G  5.0G  27% /
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  868K  191M   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
/dev/xvda16     881M   76M  744M  10% /boot
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs            96M   12K   96M   1% /run/user/1000
```

📌 **Key Point:** You can see i have used filesystem and volumes interchangeably in this context

Here's why:

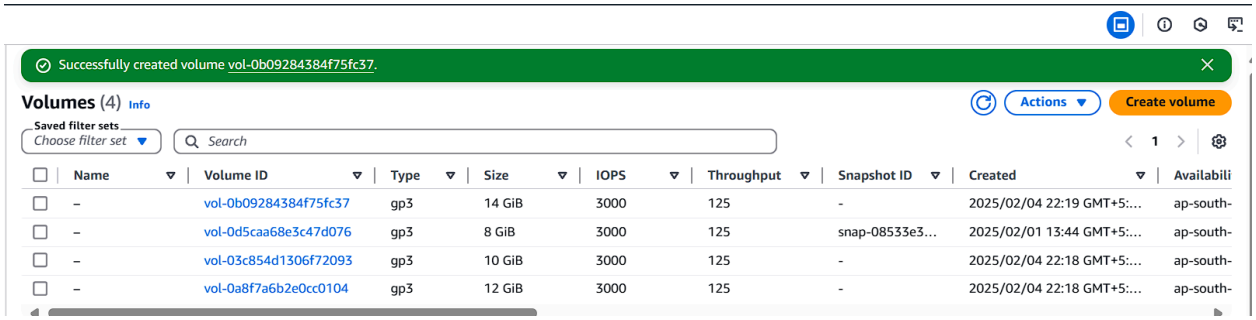
1️⃣ **Volume** refers to a logical storage unit, such as a disk or partition.

2️⃣ **File System** refers to the way data is organized and stored on that volume.

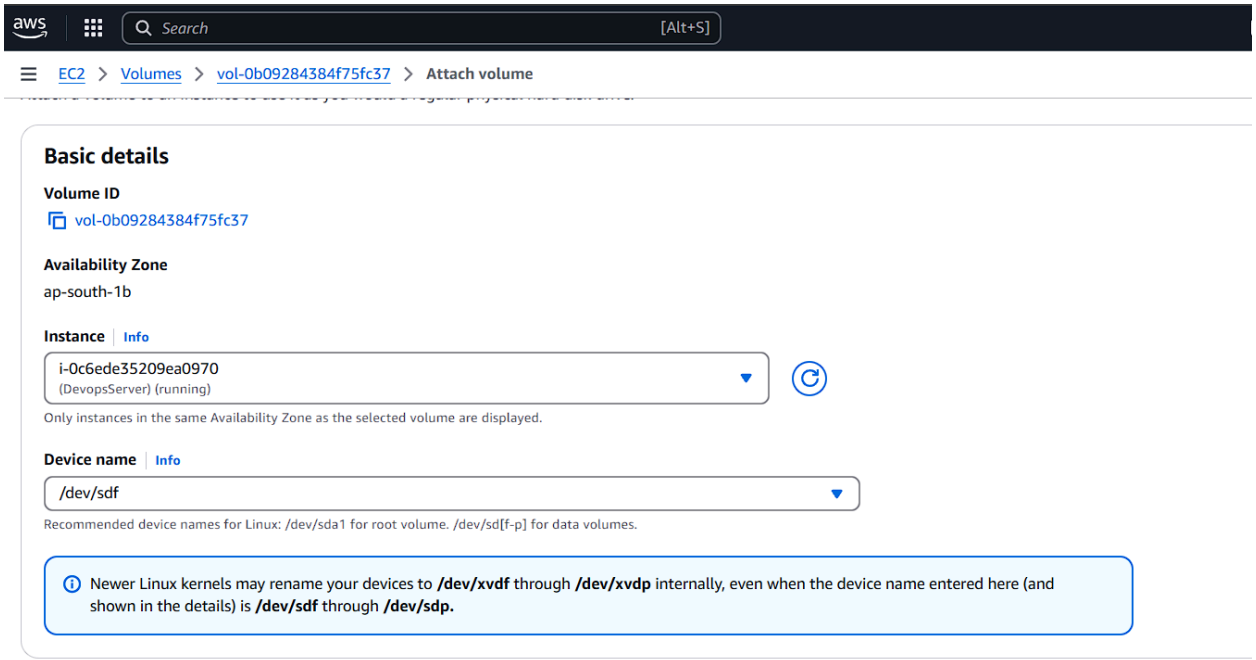
So, while "filesystem" technically refers to how data is organized on a volume, it's common to use the term "filesystem" when talking about volumes in this context, because that's what the `df-h` command reports on.

Let's roll out some additional EBS volumes and attach to our EC2 instance

I manually created volumes of sizes 10GB, 12GB, and 14GB, while the 8GB volume was automatically created when I launched the EC2 instance.

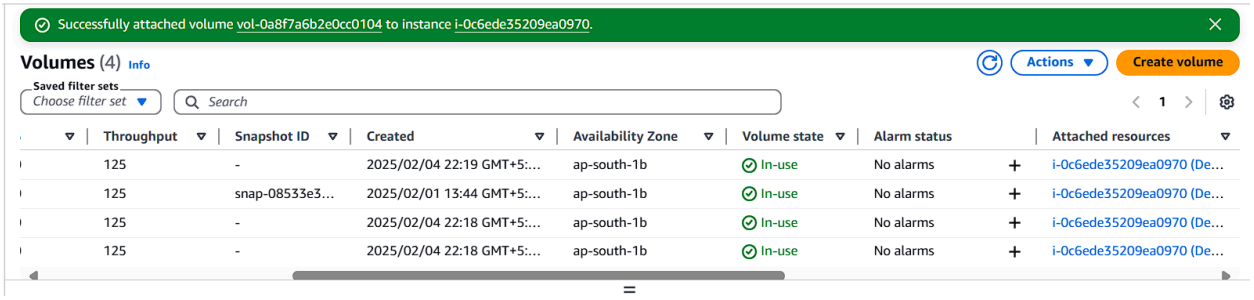


Now that we've created the volumes, we can proceed to attach them to our EC2 instance.



📌 **Key Point:** Ensure you use the correct device name as per the recommended naming conventions. I've named my 10 GB EBS volume as /dev/sdf, but Linux may automatically rename it to /dev/xvdf as mentioned .

Similarly, I've attached all the volumes, and you can do the same.



Now that we've successfully attached the volumes

let's check if they are showing up on our EC2 instance.

```
ubuntu@ip-172-31-5-208:~$ lsblk
NAME     MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0      7:0    0 26.3M  1 loop /snap/amazon-ssm-agent/988
1
loop1      7:1    0 73.9M  1 loop /snap/core22/1722
loop2      7:2    0 44.4M  1 loop /snap/snapd/23545
loop3      7:3    0 73.9M  1 loop /snap/core22/1748
xvda     202:0    0    8G  0 disk
├─xvda1  202:1    0    7G  0 part /
├─xvda14 202:14   0    4M  0 part
├─xvda15 202:15   0  106M  0 part /boot/efi
└─xvda16 259:0    0  913M  0 part /boot
xvdf     202:80   0   14G  0 disk
xvdg     202:96   0   10G  0 disk
xvdh     202:112  0   12G  0 disk
```

We previously added new volumes and named them **sdf, sdg, and sdh.** However, Linux renamed them to **xvdf, xvdg, and xvdh.** While the volumes are present as we can see , their mount points are empty because they have only been attached, not mounted yet .

Before Moving further lets go through some lvm commands

```
pvcreate - Create a physical volume (PV)
vgcreate - Create a volume group (VG)
lvcreate - Create a logical volume (LV)
lvextend - Extend LV size
lvreduce - Reduce LV size
vgextend - Add PV to VG
vgreduce - Remove PV from VG
```

```
lvremove - Remove LV
vgremove - Remove VG
pvremove - Remove PV
pvs      - gives summary of all physical volumes
vgs      - gives summary of all volume groups
lvs      - gives summary of all logical volumes
```

Become a root user

```
ubuntu@ip-172-31-5-208:~$ sudo su
root@ip-172-31-5-208:/home/ubuntu#
```

Run lvm command

```
root@ip-172-31-5-208: /home
root@ip-172-31-5-208:/home/ubuntu# lvm
lvm>
```

Now we are inside the Logical volume manager [LVM]

Let's create the Physical volume

EBS volumes, by default, are just raw devices that the operating system sees. To manage them with LVM (Logical Volume Manager), you need to initialize them as **Physical Volumes (PVs)** using the `pvcreate` command.

So your physical volumes are created -

```
ubuntu@ip-172-31-5-208:~$ sudo su
root@ip-172-31-5-208:/home/ubuntu# lvm
lvm> pvcreate /dev/xvdf /dev/xvdg /dev/xvdh
  Physical volume "/dev/xvdf" successfully created.
  Physical volume "/dev/xvdg" successfully created.
  Physical volume "/dev/xvdh" successfully created.
lvm>
```

You can see your physical volumes like this -

```
root@ip-172-31-5-208:/home/ubuntu# lvm
lvm> pvs
  PV         VG Fmt  Attr PSize  PFree
  /dev/xvdf     lvm2 ---  14.00g 14.00g
  /dev/xvdg     lvm2 ---  10.00g 10.00g
  /dev/xvdh     lvm2 ---  12.00g 12.00g
lvm>
```

Let's create the volume group

Creating a volume group with /dev/xvdf and /dev/xvdg

```
lvm > vgcreate tws_vg /dev/xvdf /dev/xvdg
      Volume group "tws_vg" successfully created
lvm>
```

you can see your volume groups like this -

```
lvm> vgs
  VG      #PV #LV #SN Attr   VSize  VFree
  tws_vg    2   0   0 wz--n- 23.99g 23.99g
lvm>
```

Now finally Let’s create the logical volume

Creating a logical volume tws_lv of 10GB .

```
root@ip-172-31-5-208: /home
root@ip-172-31-5-208:/home/ubuntu# lvm
lvm> lvcreate -L 10G -n tws_lv tws_vg
  Logical volume "tws_lv" created.
lvm>
```

You can see your logical volumes like this -

```
lvm> lvs
  LV     VG     Attr       LSize  Pool Origin Data%  Meta%
Move Log Cpy%Sync Convert
  tws_lv tws_vg -wi-a----- 10.00g

lvm>
```

View Attached Storage Devices with lsblk

```
root@ip-172-31-5-208:/home/ubuntu# lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0             7:0    0 26.3M  1 loop /snap/amazon-ssm-ag
ent/9881
loop1             7:1    0 73.9M  1 loop /snap/core22/1722
loop2             7:2    0 44.4M  1 loop /snap/snapd/23545
loop3             7:3    0 73.9M  1 loop /snap/core22/1748
xvda            202:0    0    8G  0 disk
├─xvda1         202:1    0    7G  0 part /
├─xvda14        202:14   0    4M  0 part
├─xvda15        202:15   0  106M  0 part /boot/efi
└─xvda16        259:0    0  913M  0 part /boot
xvdf            202:80   0   14G  0 disk
└─tws_vg-tws_lv 252:0    0   10G  0 lvm
xvdg            202:96   0   10G  0 disk
xvdh            202:112  0   12G  0 disk
root@ip-172-31-5-208:/home/ubuntu#
```

📌 **Key Point:** We can see that a **10GB logical volume** has been created, which is currently allocated from **xvdf**. In the future, if we create more logical volumes, they might be allocated from **xvdg** as well.

Check Disk Space Usage with df -h

```
root@ip-172-31-5-208:/home/ubuntu# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       6.8G  1.8G  4.9G  27% /
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  884K  191M   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
/dev/xvda16     881M   76M  744M  10% /boot
/dev/xvda15     105M  6.1M   99M   6% /boot/efi
tmpfs            96M   12K   96M   1% /run/user/1000
root@ip-172-31-5-208:/home/ubuntu#
```

📌 **Key Point :**But surprisingly We can't see **xvdf, xvdg, and xvdh** because they are not mounted yet. Let's go ahead and mount them.

Let’s do Mounting Now

1- Mounting Logical volume tws_lv

- **Create a Mount Point**

Before mounting, we need to create a directory where the logical volume will be accessible.

```
root@ip-172-31-5-208:/home/ubuntu# mkdir /mnt/tws_lv_mount
```

- **Format the Logical Volume**

To create a filesystem on the logical volume, we use the **`mkfs.ext4`** command.

You can format any logical volume by specifying its correct path, which follows the format:

```
syntax - mkfs.ext4 /dev/<VG_NAME>/<LV_NAME>
```

Replace **`<VG_NAME>`** and **`<LV_NAME>`** with your actual volume group and logical volume names.

```
root@ip-172-31-5-208: /home
root@ip-172-31-5-208:/home/ubuntu# mkfs.ext4 /dev/tws_vg/tws
_lv
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: c4de9e18-1d44-4d51-94c3-3ff430e88e68
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736
, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information:
done
```

- **Mount the logical volume**

```
 mount <logical_volume> <mount_point>
<logical_volume>: Path to the logical volume (e.g., /dev/tws_vg/tws_lv).
<mount_point>: Directory where the volume will be mounted (e.g.,
/mnt/tws_lv_mount).
```

```
root@ip-172-31-5-208:/home/ubuntu# mount /dev/tws_vg/tws_lv
/mnt/tws_lv_mount
root@ip-172-31-5-208:/home/ubuntu#
```

After running this command, the contents of **`/dev/tws_vg/tws_lv`** will be available inside **`/mnt/tws_lv_mount`.**

📌 **Key Point :** To confirm that the logical volume has been successfully mounted, use the **`df -h`** command:

```
root@ip-172-31-5-208: /home
root@ip-172-31-5-208:/home/ubuntu# df -h
Filesystem                  Size  Used Avail Use% Mounted on
/dev/root                   6.8G  1.8G  4.9G  27% /
tmpfs                       479M     0  479M   0% /dev/shm
tmpfs                       192M  884K  191M   1% /run
tmpfs                       5.0M     0  5.0M   0% /run/lock
/dev/xvda16                 881M   76M  744M  10% /boot
/dev/xvda15                 105M  6.1M   99M   6% /boot/efi
tmpfs                        96M   12K   96M   1% /run/user/1
000
/dev/mapper/tws_vg-tws_lv   9.8G   24K  9.3G   1% /mnt/tws_lv
_mount
root@ip-172-31-5-208:/home/ubuntu#
```

We can see our logical volume **tws_lv** successfully mounted at **/mnt/tws_lv_mount**

Now that the logical volume is mounted, the space is ready for use.

You can create new folders and files within it as needed.

```
root@ip-172-31-5-208:/home/ubuntu# cd /mnt/tws_lv_mount/
root@ip-172-31-5-208:/mnt/tws_lv_mount# mkdir newfolder
root@ip-172-31-5-208:/mnt/tws_lv_mount# touch newfile.txt
root@ip-172-31-5-208:/mnt/tws_lv_mount#
```

```
root@ip-172-31-5-208: /mnt/t
root@ip-172-31-5-208:/mnt/tws_lv_mount# ls
lost+found  newfile.txt  newfolder
root@ip-172-31-5-208:/mnt/tws_lv_mount#
```

2- Mounting AWS EBS volume disk directly

- **Create a Mount Point**

To mount an AWS EBS volume to an EC2 instance, first, you need to create a mount point directory where the volume will be attached.

```
root@ip-172-31-5-208: /
root@ip-172-31-5-208:/# mkdir /mnt/tws_diskmount
root@ip-172-31-5-208:/#
```

The first directory is our mounting point

```
root@ip-172-31-5-208:/mnt# ls
tws_diskmount  tws_lv_mount
root@ip-172-31-5-208:/mnt#
```

- **Format the AWS EBS volume disk**

To format an AWS EBS volume disk , you can use the following syntax:

```
syntax - mkfs -t ext4 <path_to_aws_ebs_volume>
```

```
root@ip-172-31-5-208:/mnt# mkfs -t ext4 /dev/xvdh
mke2fs 1.47.0 (5-Feb-2023)
/dev/xvdh contains a LVM2_member file system
Proceed anyway? (y,N) y
Creating filesystem with 3145728 4k blocks and 786432 inodes
Filesystem UUID: eaf90f5c-d132-4aa9-b783-2da61b20d57b
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736
, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information:
done

root@ip-172-31-5-208:/mnt#
```

- **Mount the AWS EBS volume disk**

Use the following command to mount the EBS volume **(`/dev/xvdh`)** to the mount point **(`/mnt/ebs_volume`)**

```
root@ip-172-31-5-208: /

root@ip-172-31-5-208:/# mount /dev/xvdh /mnt/tws_diskmount
root@ip-172-31-5-208:/#
```

📌 **Key Point :** To confirm that the logical volume has been successfully mounted, use the **`df -h`** command:

```
root@ip-172-31-5-208:/# df -h
Filesystem                 Size  Used Avail Use% Mounted on
/dev/root                  6.8G  1.8G  4.9G  27% /
tmpfs                      479M     0  479M   0% /dev/shm
tmpfs                      192M  884K  191M   1% /run
tmpfs                      5.0M     0  5.0M   0% /run/lock
/dev/xvda16                881M   76M  744M  10% /boot
/dev/xvda15                105M  6.1M   99M   6% /boot/efi
tmpfs                       96M   12K   96M   1% /run/user/1
000
/dev/mapper/tws_vg-tws_lv  9.8G   28K  9.3G   1% /mnt/tws_lv
_mount
/dev/xvdh                   12G   24K   12G   1% /mnt/tws_di
skmount
root@ip-172-31-5-208:/#
```

You can see that all mounted filesystems are listed, and **`/dev/xvdh`** (the AWS EBS volume) is mounted at **`/mnt/tws_diskmount`**. Now, you can make use of this space for storing files and directories.

Conclusion

I hope this guide helped you understand how to efficiently manage storage on AWS EC2 using LVM. From creating Physical Volumes to formatting and mounting AWS EBS volumes, we've covered it all. With this knowledge, you're now equipped to manage scalable and flexible storage effectively. Keep learning and growing!

Hare krishna 🙏